IEEE Standards

IEEE standards are technical rules and guidelines developed by the Institute of Electrical and Electronics Engineers (IEEE) to ensure **compatibility, interoperability, safety, and quality** in electrical, electronic, and computer technologies.

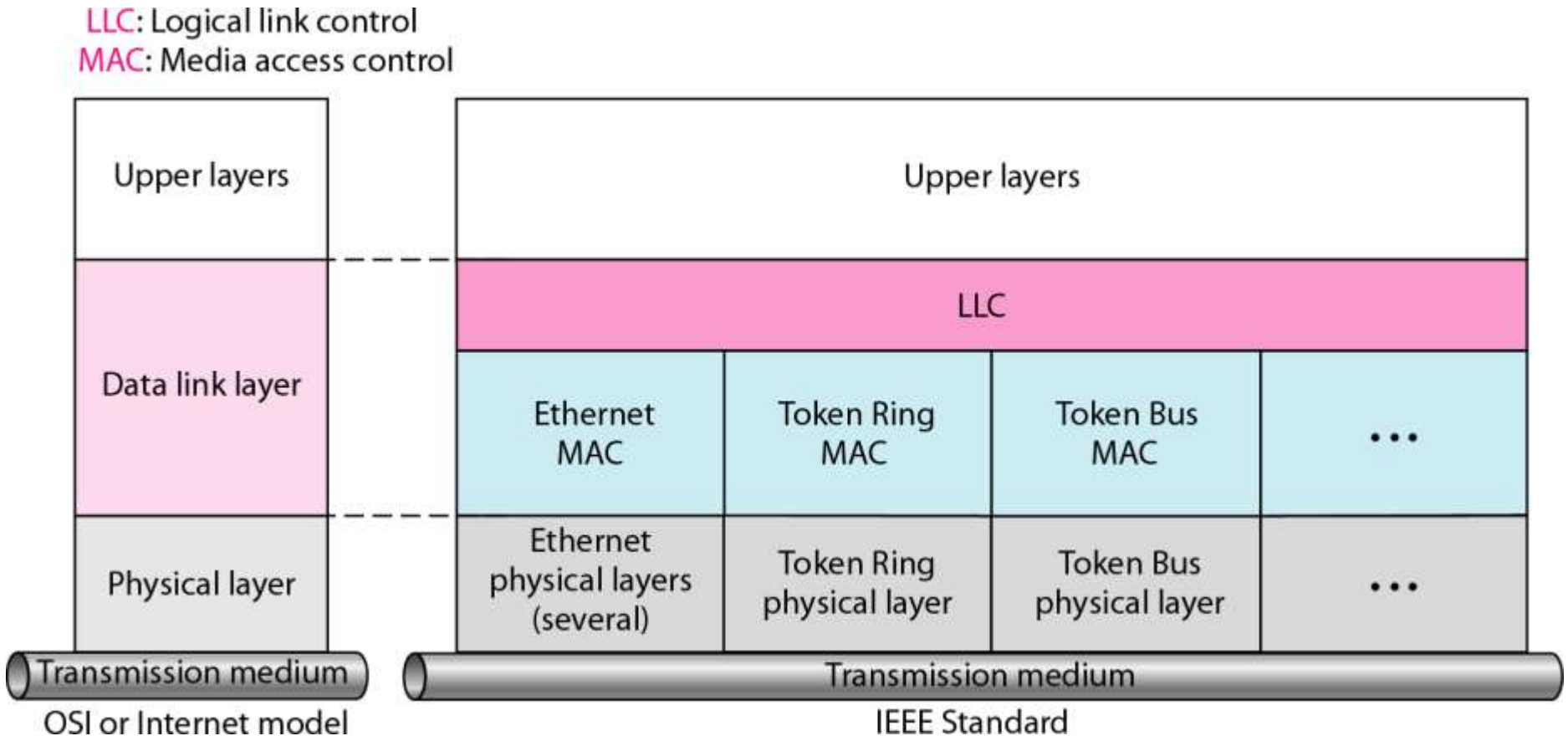
IEEE standards are internationally accepted technical specifications created by IEEE to regulate the design and operation of networking, electrical, and computing technologies.

IEEE 802 Standards

- **In 1985, the Computer Society of the IEEE (The Institute of Electrical and Electronics Engineers) started a project, called Project 802, to set standards to enable intercommunication among equipment from a variety of manufacturers.**

- **Project 802 is a way of specifying functions of the physical layer and the data link layer of major LAN protocols.**

IEEE standard for LANs



Notable IEEE Standards formats

IEEE 802	LAN/MAN
IEEE 802.1	Standards for LAN/MAN bridging and management and remote media access control (MAC) bridging.
IEEE 802.2	Standards for Logical Link Control (LLC) standards for connectivity.
IEEE 802.3	Ethernet Standards for Carrier Sense Multiple Access with Collision Detection (CSMA/CD).
IEEE 802.4	Standards for token passing bus access.
IEEE 802.5	Standards for token ring access and for communications between LANs and MANs
IEEE 802.6	Standards for information exchange between systems.
IEEE 802.7	Standards for broadband LAN cabling.
IEEE 802.8	Fiber optic connection.

IEEE 802.9	Standards for integrated services, like voice and data.
IEEE 802.10	Standards for LAN/MAN security implementations.
IEEE 802.11	Wireless Networking – "Wi-Fi".
IEEE 802.12	Standards for demand priority access method.
IEEE 802.14	Standards for cable television broadband communications.
IEEE 802.15.1	Bluetooth
IEEE 802.15.4	Wireless Sensor/Control Networks – "ZigBee"
IEEE 802.15.6	Wireless Body Area Network (BAN) – (e.g. Bluetooth low energy)
IEEE 802.16	Wireless Networking – "WiMAX"

IEEE 802.3(Wired LAN's: Ethernet)

IEEE 802.3 is the Ethernet standard developed by the Institute of Electrical and Electronics Engineers for wired LAN communication.

It defines the physical layer and MAC layer specifications and uses the CSMA/CD protocol for transmitting data over Ethernet networks.

What is Ethernet?

- **Ethernet is a family of computer networking technologies for local area networks (LANs) and metropolitan area networks (MANs).** It was commercially introduced in 1980 and first standardized in 1983 as IEEE 802.3 and has since been refined to support higher bit rates and longer link distances.
- Over time, Ethernet has largely replaced competing wired LAN technologies such as token ring, FDDI, and ARCNET.

Working Principle (CSMA/CD)

In IEEE 802.3 networks:

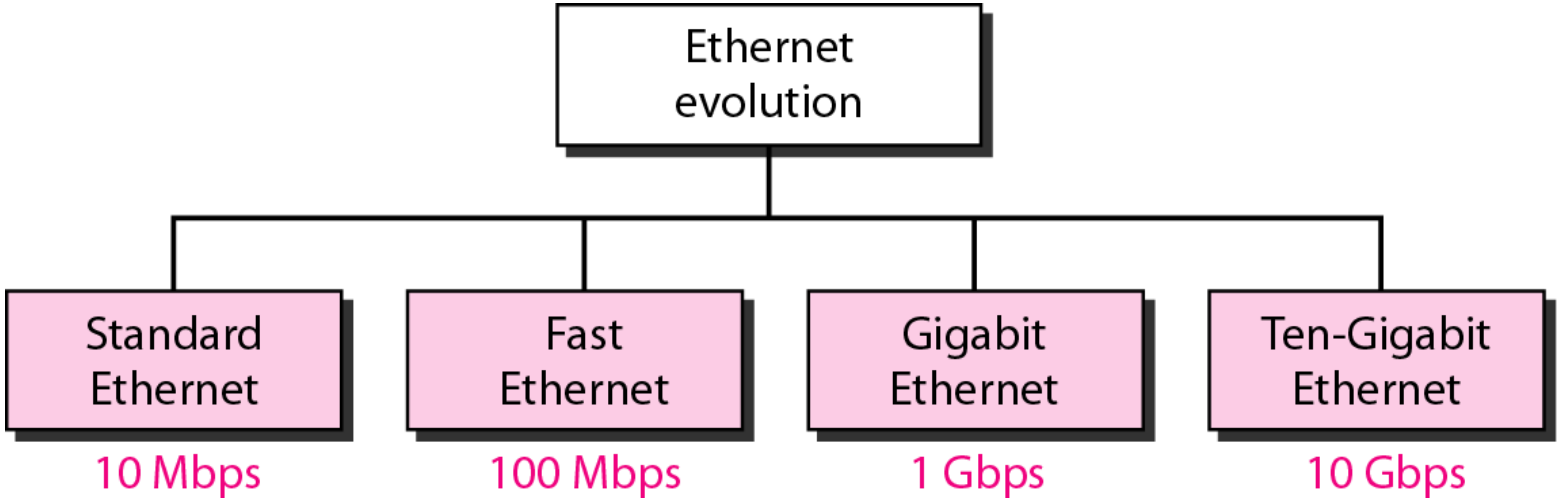
1. A device **listens to the channel** before transmitting.

2. If the channel is **idle**, it sends the frame.

3. If two devices transmit simultaneously, a **collision occurs**.

4. Collision is detected and both devices **wait a random time** before retransmitting.

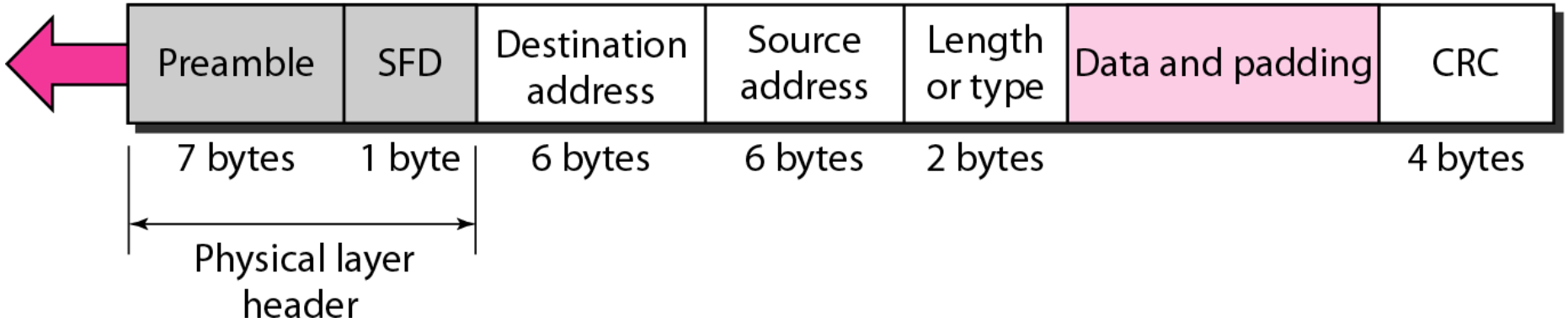
Ethernet evolution through four generations



802.3 MAC Frame Format

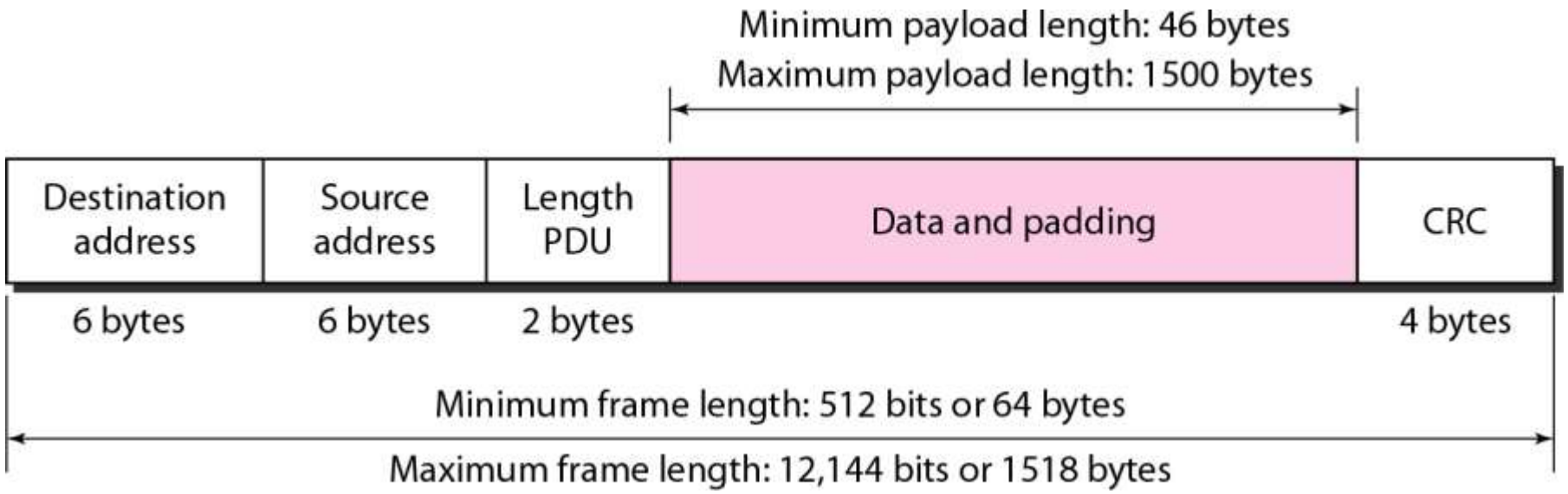
Preamble: 56 bits of alternating 1s and 0s.

SFD: Start frame delimiter, flag (10101011)



- **Preamble** – A 7-byte field providing synchronization and timing information for the receiving station.
- **Start of Frame Delimiter (SOF)** – A 1-byte field containing the pattern 10101011 that marks the beginning of the frame.
- **Destination Address** – A 6-byte field containing the MAC address of the receiving station.
- **Source Address** – A 6-byte field containing the MAC address of the sending station.
- **Length** – A 2-byte field that specifies the number of bytes in the data field.
- **Data** – A variable-sized field (46-1500 bytes) that carries the actual information from upper layers.
- **Padding** – Additional bytes added to ensure the data field meets the minimum requirement of 46 bytes.
- **Frame Check Sequence (FCS/CRC)** – A 4-byte field containing **cyclic redundancy check** information for **error detection**.

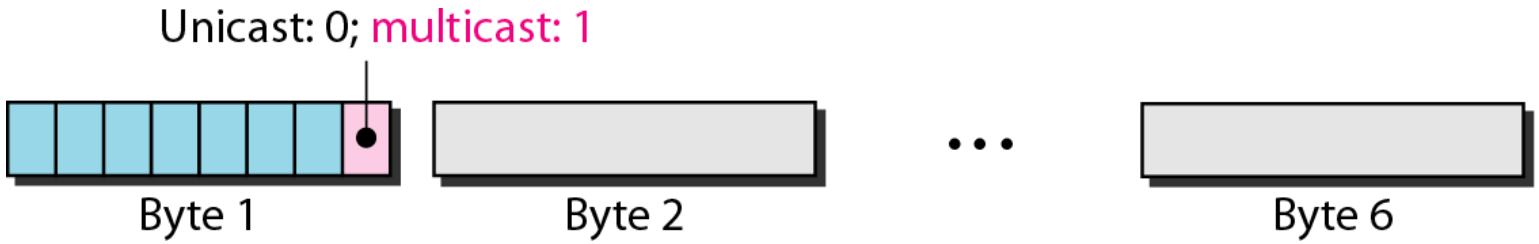
Minimum and maximum lengths



Example of an Ethernet address in hexadecimal notation

06 : 01 : 02 : 01 : 2C : 4B

6 bytes = 12 hex digits = 48 bits



- The least significant bit of the first byte defines the type of address. **If the bit is 0, the address is unicast; otherwise, it is multicast.**
- The broadcast destination address is a special case of the multicast address in which all bits are 1s.

Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A

b. 47:20:1B:2E:08:EE

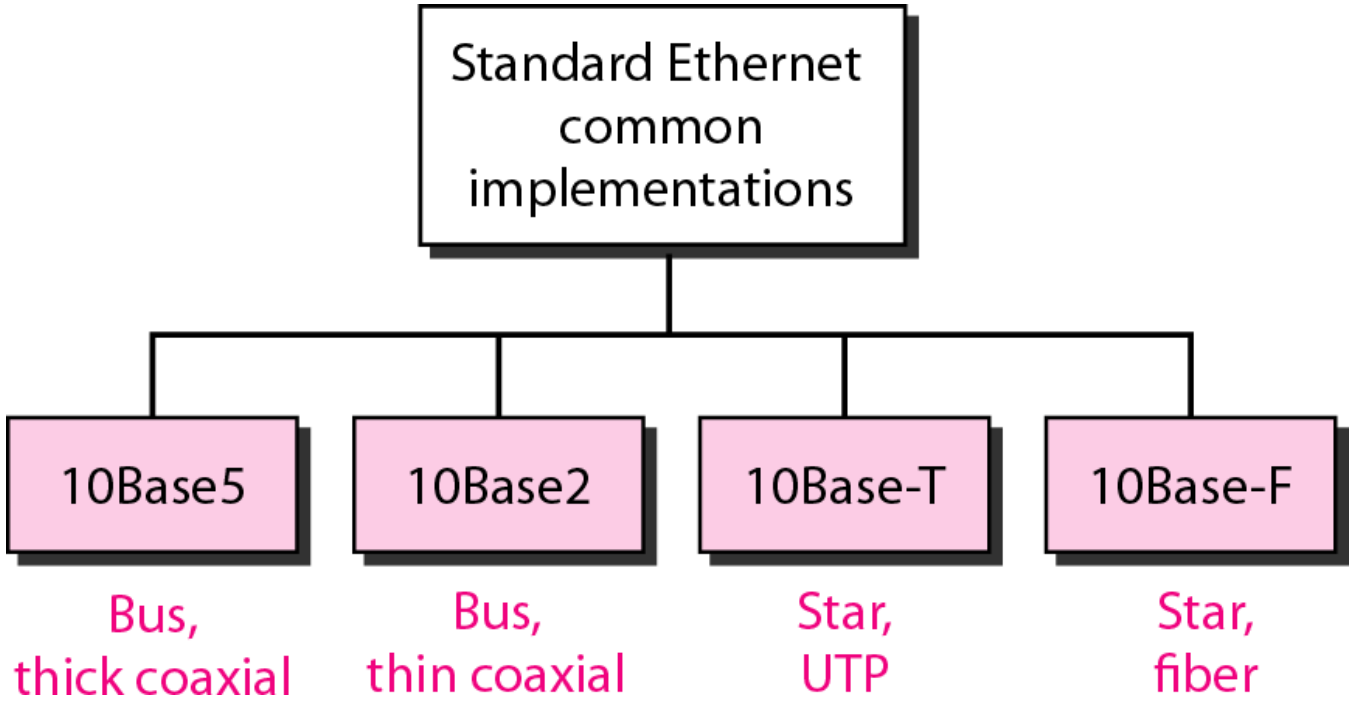
c. FF:FF:FF:FF:FF:FF

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are F's, the address is broadcast. Therefore, we have the following:

a. ***This is a unicast address because A in binary is 1010.***
b. ***This is a multicast address because 7 in binary is 0111.***
c. ***This is a broadcast address because all digits are F's.***

Categories of Standard Ethernet (802.3)

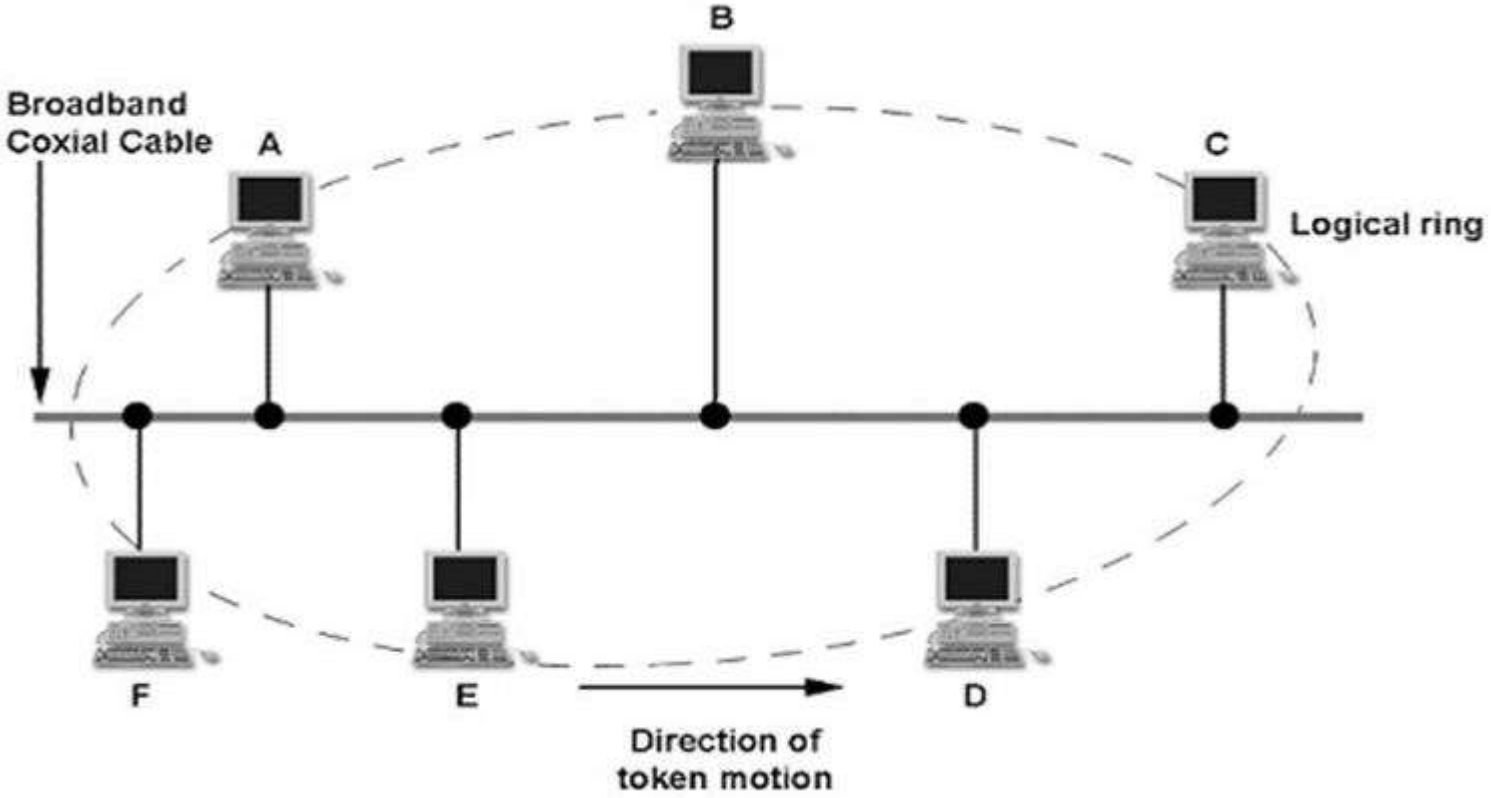


Ethernet Cabling

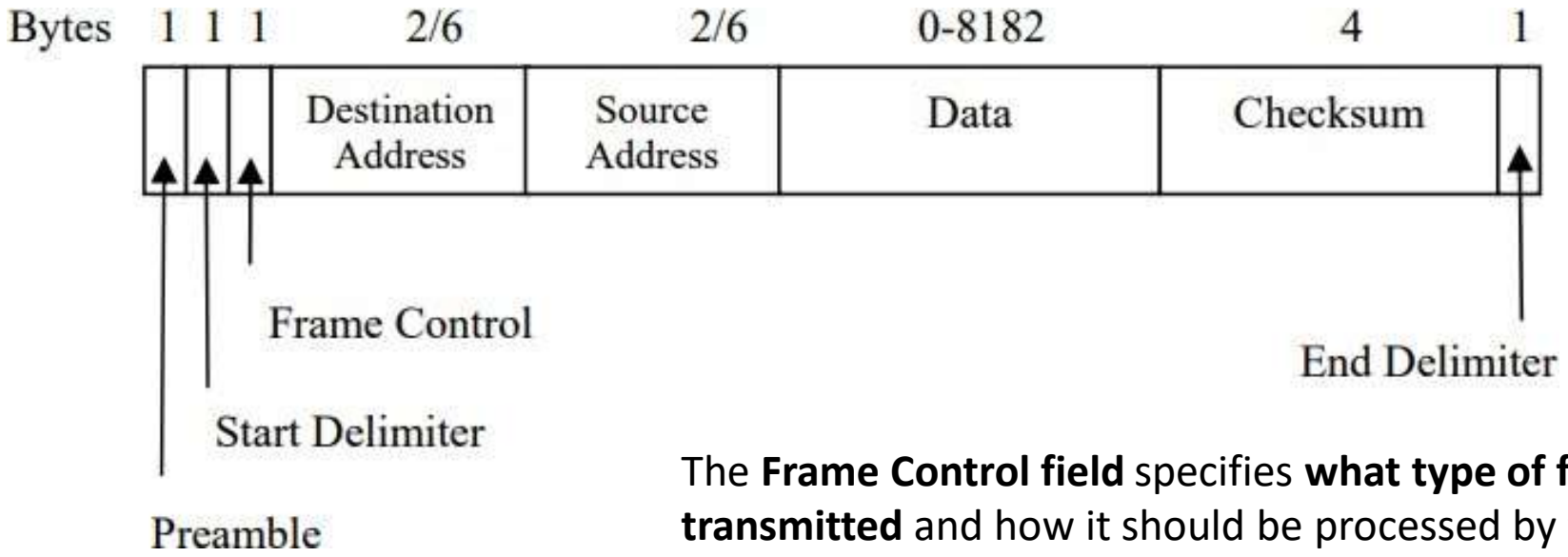
Name	Cable	Max seg. (m)	Nodes per segment	Advantages
10Base5	thick coax	**500**	**100**	The Original
10Base2	thin coax	**185**	**30**	no hub
10Base-T	twisted pair (UTP)	**100**	**1024**	cheapest
10Base-F	fiber	**2000**	**1024**	long distance

Token Bus – IEEE 802.4

- A network which implements the modified **Token Ring** protocol over a **"virtual ring"** on a coaxial cable with a bus topology.
- It is mainly used for industrial applications.



- The IEEE 802.4 is a popular standard for the bus based token passing LANs. In a token bus LAN, the physical media is a bus and a logical ring as shown in the figure is created. The token is passed from one user to other in a sequence, decided in the logical ring. **IEEE 802.4 frame** is as shown in the figure-



The **Frame Control field** specifies **what type of frame is being transmitted** and how it should be processed by the receiving stations.

It helps stations determine whether the frame is:

•a **data frame** a **control frame** a **token frame**

Control Token or Token Passing

Working Principle

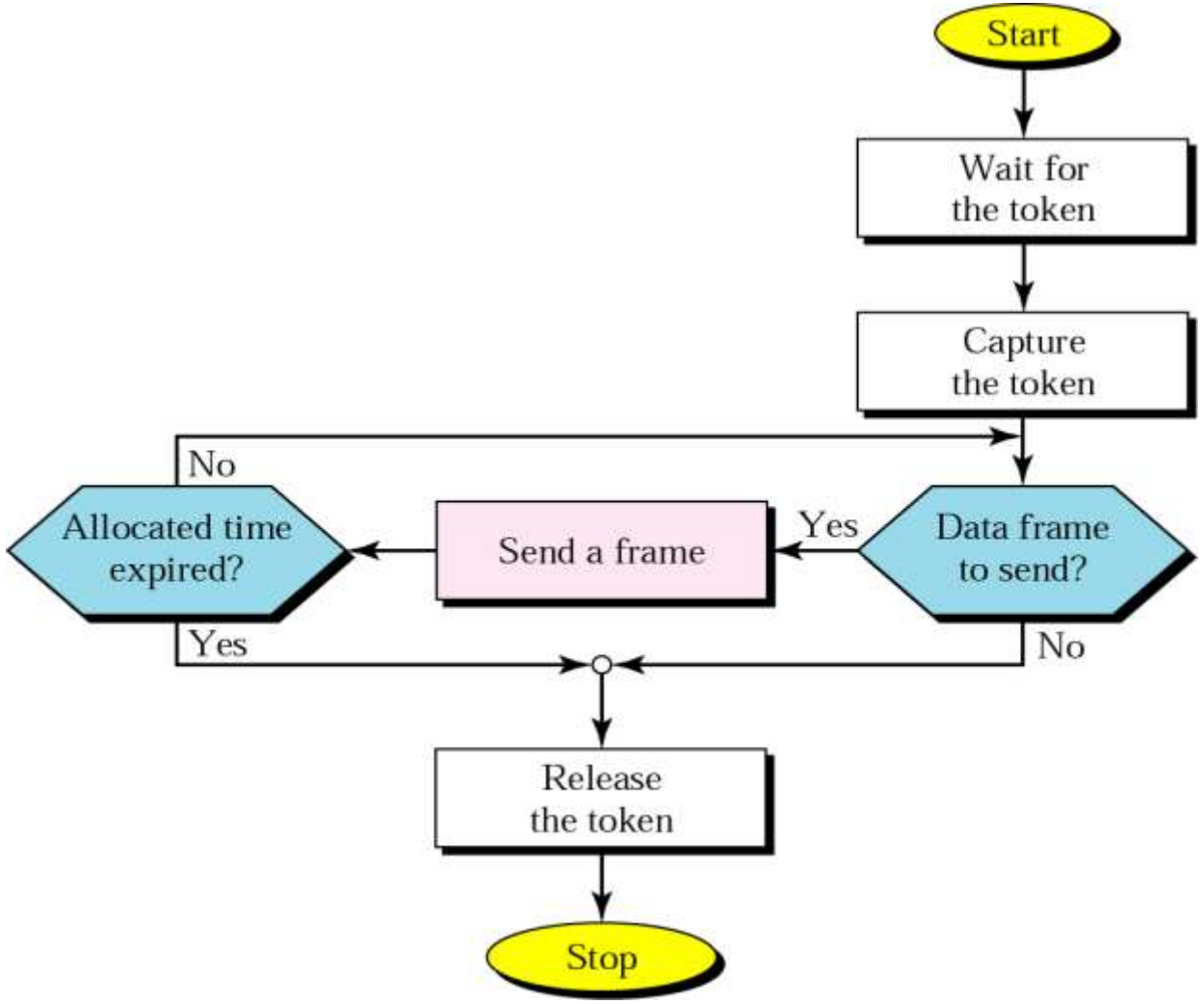
1. All devices are connected to a **single bus cable**.

2. A special control frame called a **token** circulates among the stations in a **logical ring order**.

3. Only the station holding the **token** is allowed to transmit data.

4. After sending data, the station **passes the token to the next station**.

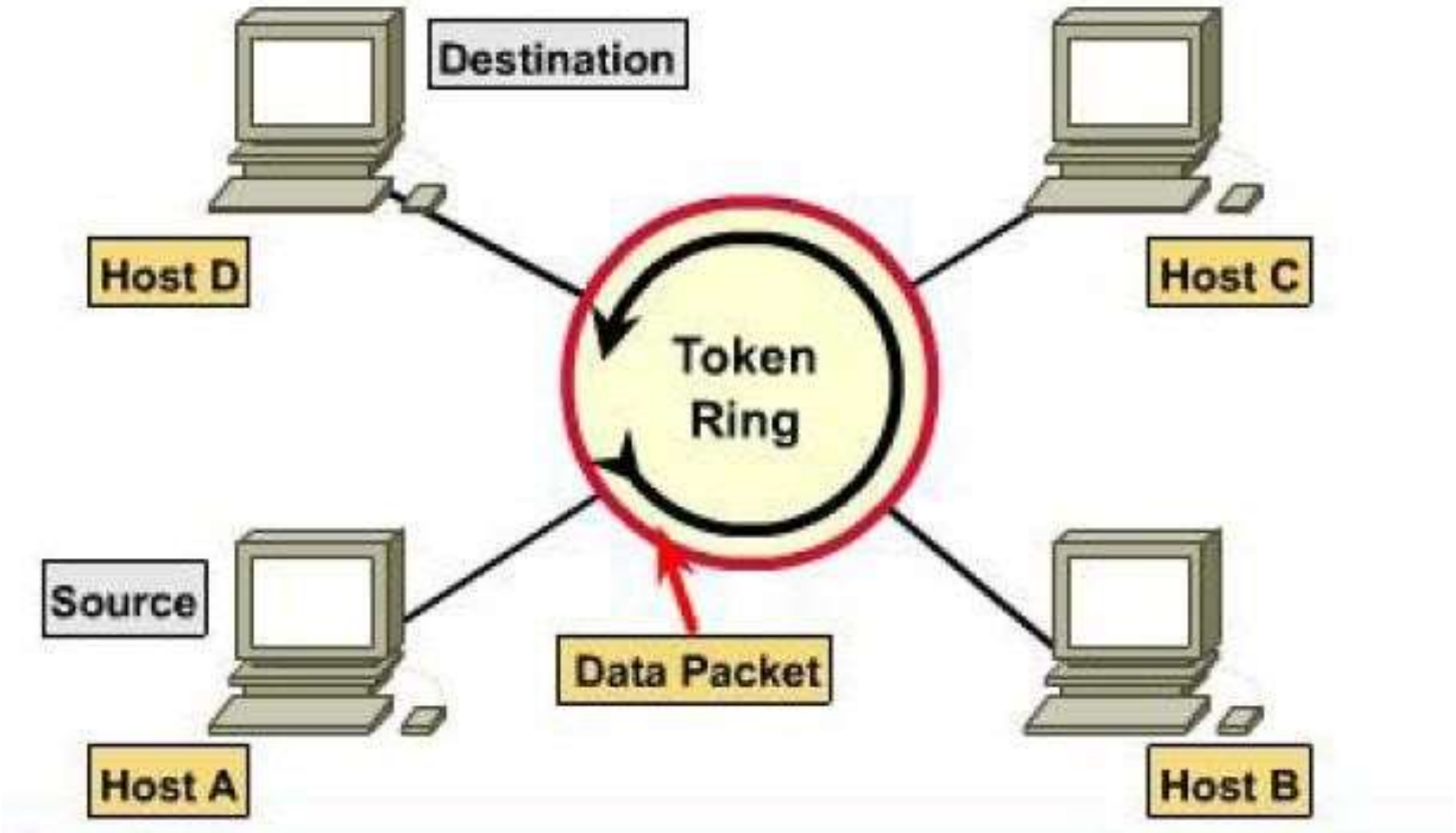
Control Token procedure



Token Ring – IEEE 802.5

- A **ring** topology network developed in the 1970s. Supported mainly by IBM.
- A LAN protocol which resides at the data link layer (**DLL**) of the **OSI** model.
- In cases of heavy traffic, the token ring network has higher throughput than ethernet due to the deterministic (non-random) nature of the medium access.
- Is used in applications in which delay when sending data must be predictable
- Is a robust network i.e. it is fault tolerant through fault management mechanisms
- Can support data rates of around 16 Mbps.
- **Typically uses twisted pair**

Token Ring – IEEE 802.5



Working Principle

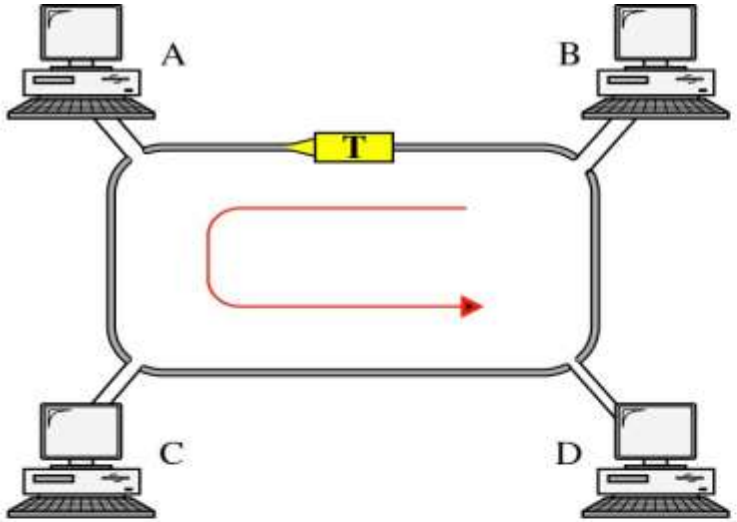
1. All computers are connected in a **logical ring**.

2. A special frame called a **token** continuously circulates around the ring.

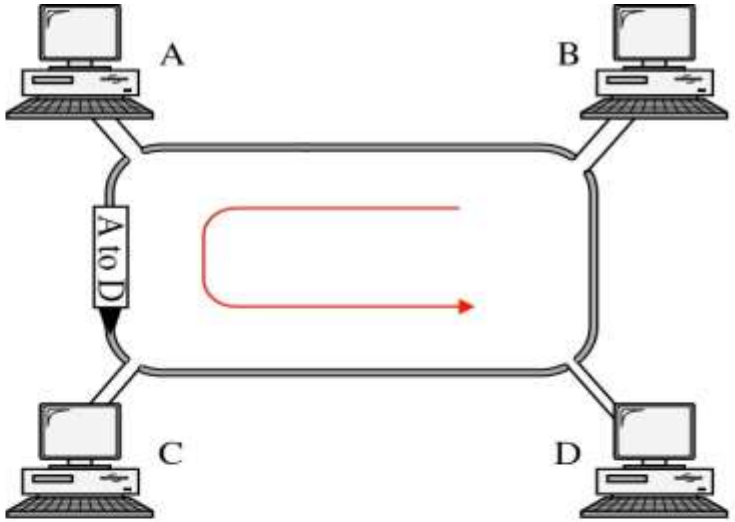
3. When a station receives the **free token**, it can transmit data.

4. After transmission, the token is **released back to the network** for the next station.

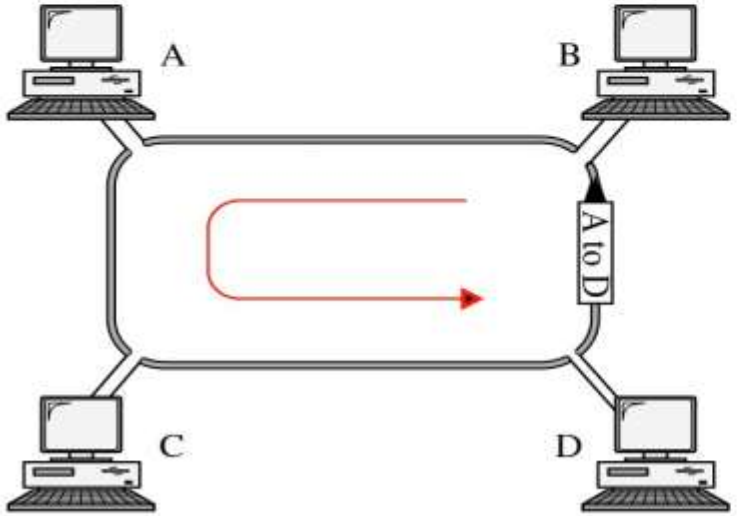
This ensures that **only one device transmits at a time**, avoiding collisions.



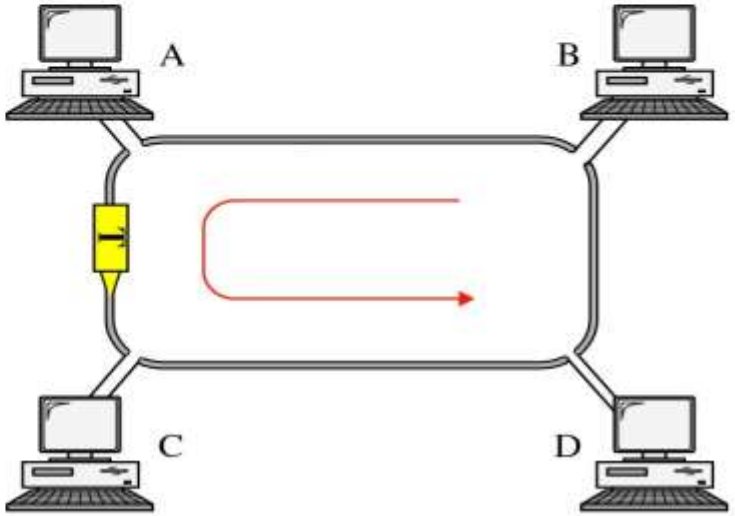
a. Token is traveling along the ring.



b. Station A captures the token and sends its data to D.



c. Station D copies the frame and sends the data back to the ring.



d. Station A receives the frame and releases the token.

Token Ring frame formats

Data or command frame

SD	AC	FC	DA	SA	PDU from LLC (IEEE 802.2)	CRC	ED	FS
8 bits	8 bits	8 bits	48 bits	48 bits	up to 18200x8 bits	32 bits	8 bits	8 bits

Token

SD	AC	ED
8 bits	8 bits	8 bits

Abort frame

SD	ED
8 bits	8 bits

SD: Starting delimiter

AC: Access control

FC: Frame control

DA: Destination address

SA: Source address

PDU: data

CRC: check sum

ED: End delimiter

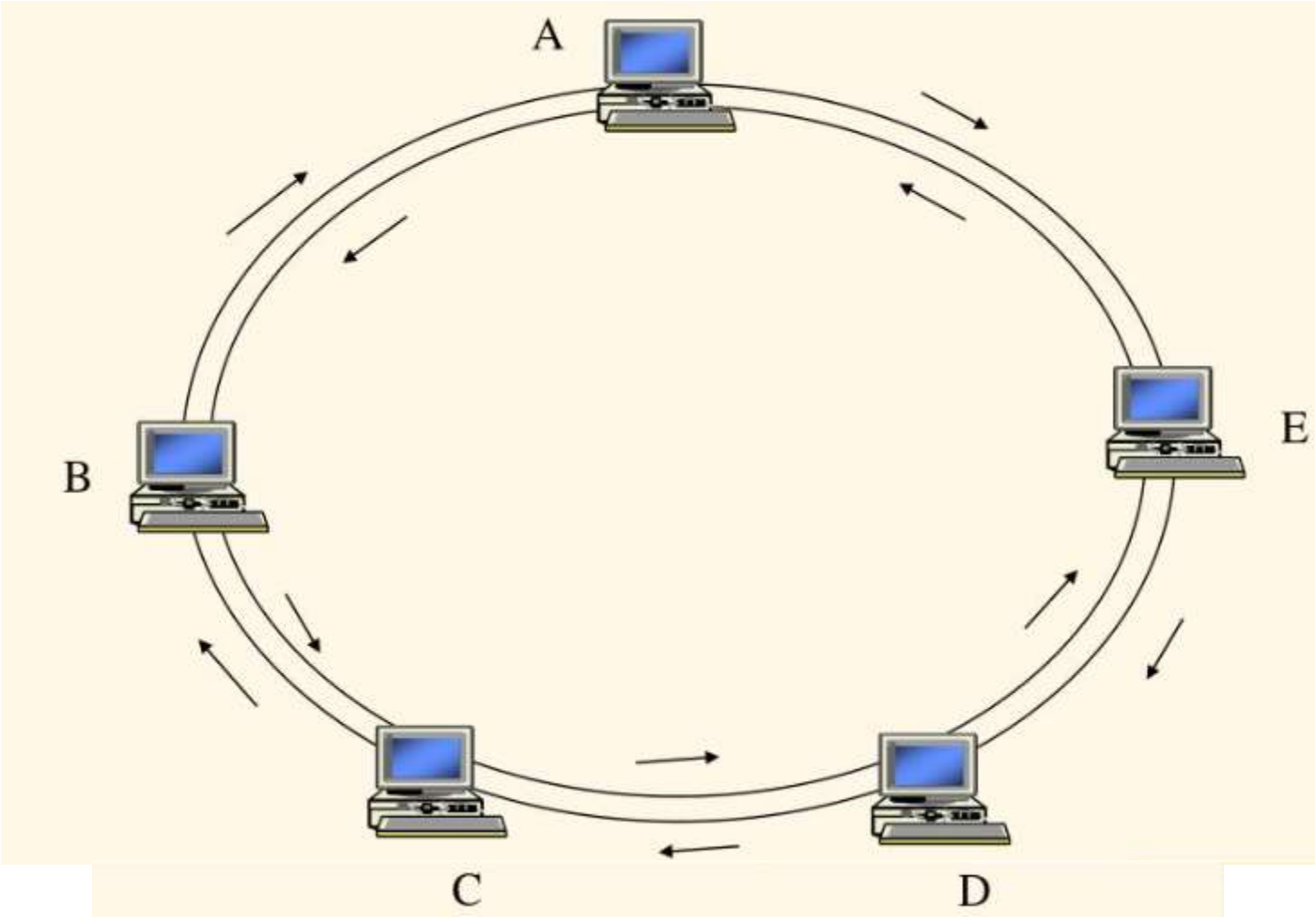
FS: Frame status

The **Access Control field** controls **how stations access the token ring network**.It determines **priority levels, token control, and reservation for transmission**.

The **Frame Status field** indicates whether the **destination station has received and copied the frame successfully**.

FDDI (Fiber Distributed Data Interface)

- FDDI is a standard developed by the American National Standards Institute (ANSI) for transmitting data on optical fibers
- Supports transmission rates of up to 200 Mbps
- Uses a dual ring
  - First ring used to carry data at 100 Mbps
  - Second ring used for primary backup in case first ring fails
  - If no backup is needed, second ring can also carry data, increasing the data rate up to 200 Mbps
- Supports up to 1000 nodes
- Has a range of up to 200 km


A
B
C
D
E

FDDI (Fiber Distributed Data Interface) is a **high-speed network standard** used for **data transmission over fiber optic cables** in Local Area Networks (LAN) and Metropolitan Area Networks (MAN).
It was developed by the American National Standards Institute for reliable and fast network communication.

Working Principle

1. A **token circulates** around the ring.
2. A station can transmit data **only when it receives the token**.
3. After transmission, the token is **passed to the next station**.
4. If a link fails, the **dual ring reconfigures automatically**.

Advantages:

- **High speed communication**
- **Reliable network** due to dual ring
- **Long distance coverage**
- **Low error rate** due to fiber optics

Differences between 802.5 and FDDI

- Token Ring
- Shielded twisted pair
- 4, 16 Mbps
- No reliability specified
- Differential Manchester
- Centralized clock
- Priority and Reservation bits
- New token *after receive*

FDDI

- Optical Fiber
- 100 Mbps
- Reliability specified (dual ring)
- 4B/5B encoding
- Distributed clocking
- Timed Token Rotation Time
- New token *after transmit*